

Mapping Serotonergic Dynamics using Drug-Modulated Molecular Connectivity

Reviewed Preprint

v2 • March 18, 2025

Revised by authors

Reviewed Preprint

v1 • July 24, 2024

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eLife Assessment

This **important** paper on measuring molecular connectivity using combined serotonin PET and resting-state fMRI provides both novel methods for studying the brain as well as insights into the effects of ecstasy administration. The methods are **convincing**, with the high anaesthetic dose used likely limiting network activity.

<https://doi.org/10.7554/eLife.97864.2.sa3>

Abstract

Understanding the complex workings of the brain is one of the most significant challenges in neuroscience, providing insights into normal brain function, neurological diseases, and the effects of potential therapeutics. A major challenge in this field lies in the limitations of traditional brain imaging techniques, which often capture only fragments of the complex puzzle of brain function. Our research employs a novel approach termed “molecular connectivity” (MC), which combines the strengths of various imaging methods to provide a comprehensive view of how specific molecules, such as the serotonin transporter, interact across different brain regions and influence brain function.

This innovative technique bridges the gap between functional magnetic resonance imaging (fMRI), known for its ability to monitor brain activity by tracking blood flow, and positron emission tomography (PET), which visualizes specific molecular changes. By integrating these methods, we can better understand how drugs influence brain function. Our study focuses on the application of dynamic [^{11}C]DASB PET scans to map the distribution of serotonin transporters, key players in regulating mood and emotions and examines how these transporters are altered following exposure to methylenedioxymethamphetamine (MDMA), which is commonly known as ecstasy.

Through a detailed comparison of MCs with traditional measures of brain connectivity, we reveal significant patterns that closely align with physiological changes. Our results revealed clear changes in molecular connectivity after a single dose of MDMA, establishing a direct

link between the effects of drugs on serotonin transporter occupancy and changes in the functional brain network.

This work offers a novel methodology for the in-depth study of brain function at the molecular level and opens new pathways for understanding how drugs modulate brain activity.

Introduction

The field of neuroscience has experienced remarkable advancements over the past decade, specifically propelled by the advent of innovative imaging techniques. The simultaneous application of positron emission tomography (PET) and magnetic resonance imaging (MRI) has revolutionized our ability to assess brain function across various physiological dimensions [1, 2]. This is especially true in the context of drug effect evaluations. Combining PET with functional MRI (fMRI) has opened many investigative avenues. Techniques such as resting-state functional connectivity (rs-FC) [3] or pharmacological MRI (phMRI) [4] are now complemented by the ability of PET to quantify molecular changes, including alterations in receptor and transporter availability [5].

While fMRI provides high spatial and temporal resolution, the interpretation of its readout necessitates caution. The blood-oxygen-level-dependent (BOLD) signal in fMRI indirectly captures neuronal changes through neurovascular coupling [6], revealing only the hemodynamic consequences of molecular-level drug effects. The integration of simultaneous PET acquisition can bridge this interpretative gap by offering essential molecular insights, particularly regarding transporter or receptor alterations. Typically, PET, used independently or simultaneously with fMRI, has been used mainly in pharmacological studies to illustrate quantitative shifts in neuroreceptor or transporter availability [7]. Several studies have also explored the interregional coherence of PET tracer signals [8–10], an approach akin to fMRI-derived rs-FC. While subject-level metabolic connectivity using [^{18}F]FDG-PET has been established through the temporal correlation of regional PET signals [1, 8], studies employing transporter or receptor tracers have focused predominantly on interregional binding coherence across subjects using static scans [9, 10]. The concept and feasibility of molecular connectivity (MC) [11, 12] through the temporal correlation of the dynamic binding potentials of transporter or receptor tracers has yet to be explored.

In our study, we investigated the feasibility of deriving MCs from dynamic [^{11}C]DASB-PET scans acquired simultaneously with fMRI in rats. We divided the rats into two cohorts, a baseline group and a pharmacological application group, which were exposed to 3,4-methylenedioxymetamphetamine (MDMA). The baseline cohort assessed the feasibility of the novel methodology, contrasting it with traditional fMRI-derived rs-FC with a specific focus on temporal stability. We postulated that dynamic [^{11}C]DASB PET temporal fluctuations could be harnessed for connectivity data like that used for hemodynamic rs-FC using seed-based and independent component analysis (ICA).

The second cohort, subjected to an MDMA challenge, allowed us to evaluate the utility of our novel approach. We aimed to outline the effects of MDMA by integrating the innovative MC concept with established analysis techniques. Our primary objective in this research was to elucidate the potential of PET-derived MCs in conjunction with simultaneous PET/fMRI, exploring the avenues this methodology could open for future diagnostic and drug development studies.

Materials and Methods

Our study reevaluates two datasets previously published by our group [13, 14] to explore FC and MC simultaneously at baseline [14] and following MDMA administration [13]. Please refer to these earlier publications for detailed descriptions of the animal handling methods, experimental setups, and data acquisition procedures.

Animals

A total of 41 male Lewis rats were obtained from Charles River Laboratories (Sulzfeld, Germany). Thirty rats (354 ± 37 g, corresponding to 11 weeks of age) underwent baseline [^{11}C]DASB PET/fMRI scans without any pharmacological intervention, whereas 11 rats (365 ± 19 g, corresponding to 11 weeks of age) underwent [^{11}C]DASB PET/fMRI scans, including an acute MDMA challenge. All experiments were in compliance with German federal regulations for experimental animals and received approval from the Regierungspräsidium Tübingen.

Simultaneous PET/fMRI Experiments

The rats were subjected to simultaneous PET/fMRI experiments involving 1.3% isoflurane anesthesia, tail vein catheterization, positioning on a temperature-controlled bed, and monitoring of vital signs. The scans were performed using a 7T small-animal ClinScan MRI scanner (Bruker Biospin, Ettlingen, Germany) with a custom-developed PET insert [15]. For [^{11}C]DASB PET, we employed a bolus plus constant infusion protocol ($k_{\text{bol}} = 38.7$ min) and reconstructed the scans in 1-minute time frames. The scanning protocol and sequence parameters are outlined in detail in our previous publication [13]. The MDMA cohort received a pharmacological MDMA challenge of 3.2 mg/kg intravenously 40 minutes after tracer injection.

Data Preprocessing and Analysis

Data preprocessing followed established protocols, including steps such as realignment, mask creation, coregistration, spatial normalization, signal cleaning, and spatial smoothing, as detailed in our previous work [13]. For the MDMA dataset, PET scans were analyzed for early and late effects post-challenge using the general linear model (GLM) available in SPM. For both datasets, the baseline was defined as 30 to 40 minutes after the start of the scan. For the fMRI data, a first-level analysis was applied to the individual scans without a high-pass filter (the filter was set to “Inf”). Statistical parametric maps were generated after GLM parameter estimation using contrast vectors. The group-level analysis involved a one-sample t-test on the subject-level statistical parametric maps ($p < 0.05$, one-sided, familywise error/FWE adjusted).

Static PET scans were generated by summing dynamic frames over defined periods for 10-minute periods after the MDMA challenge (50-60 minutes to investigate early effects and 70-80 minutes to investigate late MDMA effects). Two-sample t-maps were calculated between the normalized [^{11}C]DASB uptakes of (1) the baseline scan period and the early effect period and (2) the early effect period and the late effect period ($p < 0.05$, FWE-adjusted).

All group-level t-maps were subjected to voxelwise signal quantification to determine the regional contributions of 48 regions selected according to the Schiffer atlas [16]. The average t-scores and standard deviations of all voxels were calculated.

Functional Connectivity Analysis

FC was determined using a seed-based analysis approach. The mean time series of the preprocessed BOLD-fMRI signals for each dataset across all regions (refer to *Supplementary Table 1* for the list of regions) were extracted using the SPM toolbox Marseille Boîte À Région d'Intérêt (MarsBaR). Pairwise Pearson's correlation coefficients were calculated between each pair of mean regional time series for every dataset. The Pearson's r coefficients were converted into z values using Fischer's transformation for group-level analysis. Fischer's z -transformed correlation coefficients were then used to generate mean correlation matrices for both cohorts [17].

Molecular connectivity analysis

The mean [^{11}C]DASB signal from the preprocessed PET datasets was extracted from the designed regions, including the 48 regions used for fMRI data analysis and the cerebellum using MarsBaR. Binding potentials were calculated framewise for all dynamic PET scans using the DVR-1 (Equation 1) to generate regional BP_{ND} values with the cerebellar gray matter as a reference region, which our earlier studies demonstrated to be the most appropriate for this tracer in rats [18, 19]:

$$\text{BP}_{\text{ND}} = \frac{V_T - V_{\text{ND}}}{V_{\text{ND}}} = \frac{V_T}{V_{\text{ND}}} - 1 = \text{DVR} - 1 \text{ (Eq. 1)}$$

where:

- BP_{ND} is the binding potential
- V_T is the total volume of distribution
- V_{ND} is the volume of distribution in a reference tissue
- DVR is the relative volume of distribution

To calculate the MC, we discarded the first 20 minutes of every scan, which were dominated by perfusion effects, and applied a detrending approach to the remaining 60 minutes to obtain temporally stable values (for further details, please refer to the Supplementary Methods and Supplementary Figure 1). The BP_{ND} time courses were then used to calculate the MC as described above for fMRI: ROI-to-ROI subject-level correlation matrices between all regional time courses were generated, and z -transformed correlation coefficients were used to calculate mean correlation matrices.

Independent Component Analysis

Group ICA (GIFT toolbox, MIALAB, University of New Mexico, Albuquerque, NM, USA) was used for ICA in the baseline group. The fMRI and PET preprocessed datasets were investigated between 30 and 80 minutes after the start of data acquisition. For fMRI, we selected ten independent components, while we started with two components for PET and increased the number to ten components to thoroughly dissect the varying components within the signal. The components were thresholded at a z -value ≥ 1.96 (p -value ≤ 0.05) [20], and average z -scores and standard deviations were calculated for each component. The physiological significance of the components was further explored by contrasting them with the regional [^{11}C]DASB changes induced by MDMA. Accordingly, the z scores of the independent components generated in the baseline cohort were correlated with the early and late regional [^{11}C]DASB changes induced by MDMA, measured using t scores.

Results

Comparability of MC and FC in spatial contexts and over time

We first aimed to evaluate whether MCs align spatially with FC, possess similar graph theory properties and provide consistent temporal readouts throughout the scan duration in the baseline group (**Figure 1** [↗](#)).

We found a moderate but significant correlation between the edge-level MC and FC ($r = 0.51$, $p < 0.001$; **Figure 1A and B** [↗](#)). Furthermore, both connectomes revealed small-world properties at the group level, with coefficients higher than 1 (**Figure 1C** [↗](#)). At the subject level, three molecular and four functional connectomes fell below the threshold of 1 for the small-world coefficient. Significant consistency was observed between early and late scan-derived connectomes (**Figure 1D** [↗](#) for FC, G for MC), with FC having a slight edge (**Figure 1E** [↗](#), $r = 0.96$) over MC (**Figure 1H** [↗](#), $r = 0.8$). While both FC and MC maintained steady correlation intensities, there was a negligible decline over the scan duration (**Figure 1F** [↗](#) and **1I** [↗](#)).

Deciphering the spatial characteristics of FC and MC using ICA

After establishing the feasibility of obtaining temporally stable readouts using the ROI-to-ROI approach, we employed a data-driven approach using ICA to compare the spatial characteristics of FC and MC (**Figure 2** [↗](#)).

We extracted ten group ICs from the fMRI data (**Figure 2A** [↗](#)), revealing known canonical resting-state networks, such as the posterior default-mode-like network (IC1-5, red), sensorimotor networks (IC1-5, green and purple), the anterior default-mode-like network (IC6-10, yellow) and the visual network (IC6-10, red). Given the unpredictability of the number of ICs suitable for MC IC extraction, we started with two components (**Figure 2B** [↗](#)). IC1 (orange) comprised both subcortical and cortical anterior brain regions, including the nucleus accumbens, amygdala, cingulate cortex, caudate putamen, orbitofrontal cortex and medial prefrontal cortex, whereas IC2 (blue) primarily received contributions from deeper posterior areas, such as the midbrain, thalamus, hypothalamus, periaqueductal gray cortex and medulla. Interestingly, when we extracted 10 independent components to mimic the number of components used for the FC data, the initial anterior component split into four different ICs, and the initial posterior IC split into three different ICs. Relatively clear spatial segregation can be observed for the newly formed ICs, for instance, the three posterior components extracted from specific regions (green, mainly from the medulla; red, from the hypothalamus; and part of the midbrain, blue, from the midbrain and thalamus).

MDMA-induced changes in ICA-derived molecular connectivity

Next, we aimed to explore the relationship between molecular changes in SERT availability and the molecular connectome derived from the ICA induced by acute MDMA administration (**Figure 3** [↗](#)).

Two ICs extracted from the MC showed good overlap with regions associated with the SERT salience network (IC1) and the SERT subcortical network, comprising regions with intrinsically high SERT densities (IC2). Therefore, we defined the regions contributing strongly to IC2 as the SERT salience network (CPu, Cg, NAc, Amyg, Ins, mPFC) and those with strong signals in IC1 as the SERT subcortical network (VTA, Th, MB, PAG, Hypo). Interestingly, MDMA induced immediate strong decreases in all SERT subcortical network regions, and salience areas exhibited a delay of approximately 10 minutes (**Figure 3A** [↗](#)). Voxel-level analysis revealed clear spatial overlaps between early MDMA-responsive regions and those from the posterior IC, with delayed regions mirroring the anterior IC reminiscent of the SERT salience network (**Figure 3B** [↗](#)). To quantify the striking spatial similarity between the baseline independent components and the spatiotemporal

Connectome comparison

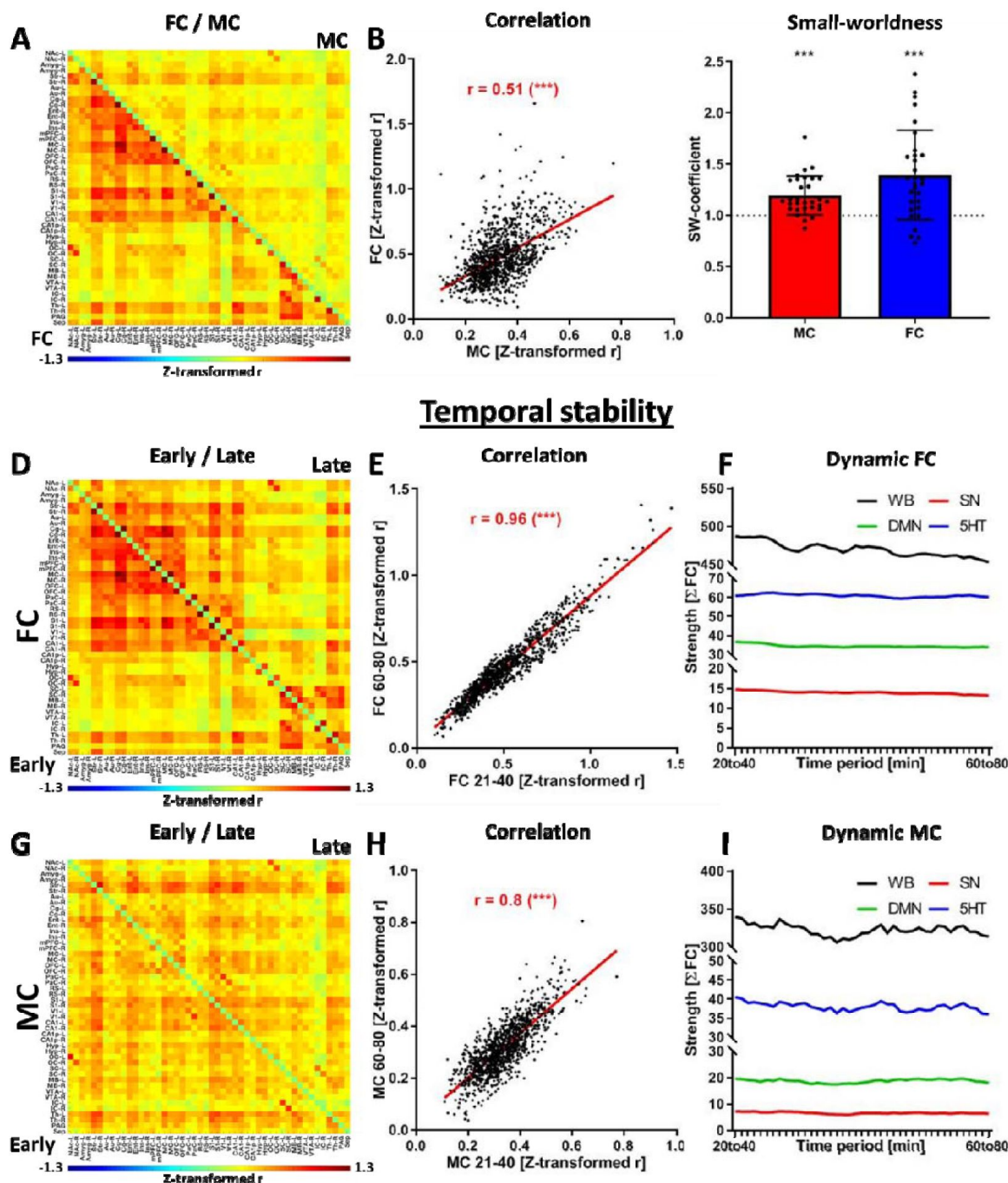


Figure 1

Evaluation of the seed-based MC.

(A) Correlation matrix indicating whole-brain FC (beneath the diagonal) and MC (above the diagonal). Correlations not significant with multiple comparison corrections were set to zero ($p < 0.05$, FWE correction). (B) Scatter plot and correlation between MC and FC edges. (C) Small-world coefficients for all subjects and group-level one-sample t-test against the value of 1 ($SW > 1$ indicates small-world properties). Comparison of (D) FC and (G) MC early (20-40 minutes after the start of the scan, below the diagonal) and late (60-80 minutes after the start of the scan, above the diagonal). The similarities of early and late readouts were quantified for both (E) FC and (H) MC. The temporal stability of both (F) FC and (I) MC was evaluated using a sliding window approach, including 20-minute windows between 20 and 80 minutes after the start of the scan. Abbreviations: FC = fMRI-derived hemodynamic functional connectivity, MC = [^{11}C]DASB PET-derived molecular connectivity.

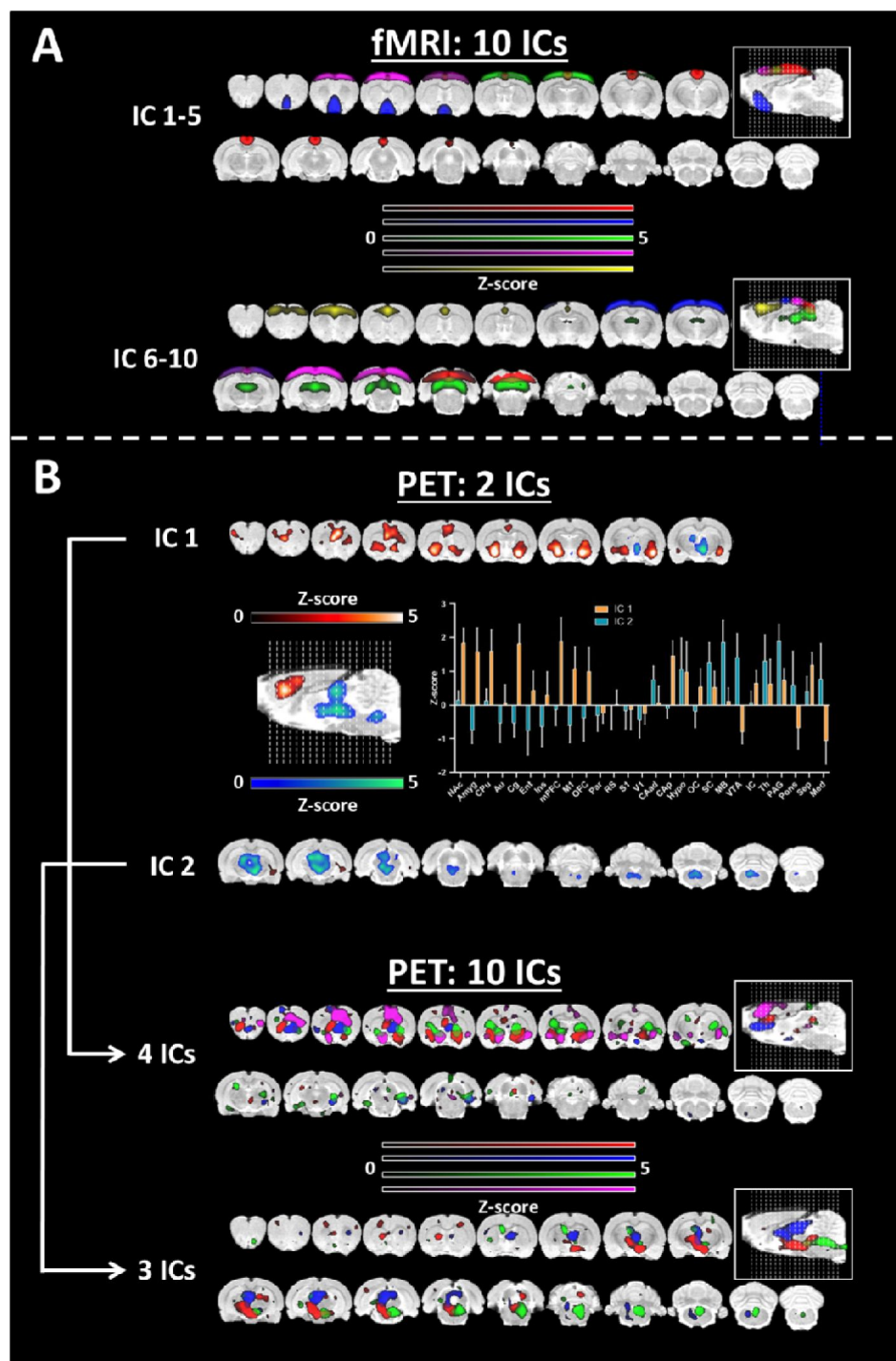


Figure 2

Group independent component analysis for FC and MC.

(A) ICA was performed over 10 components for fMRI. (B) ICA performed over 2 components for [^{11}C]DASB PET and regional quantification of the two derived components. The ICA was repeated over 10 components. Four and three components showed good overlap with the two components defined above. All the components were thresholded at $z > 1.96$ ($p \leq 0.05$). Abbreviations: FC = fMRI-derived hemodynamic functional connectivity, MC = [^{11}C]DASB PET-derived molecular connectivity.

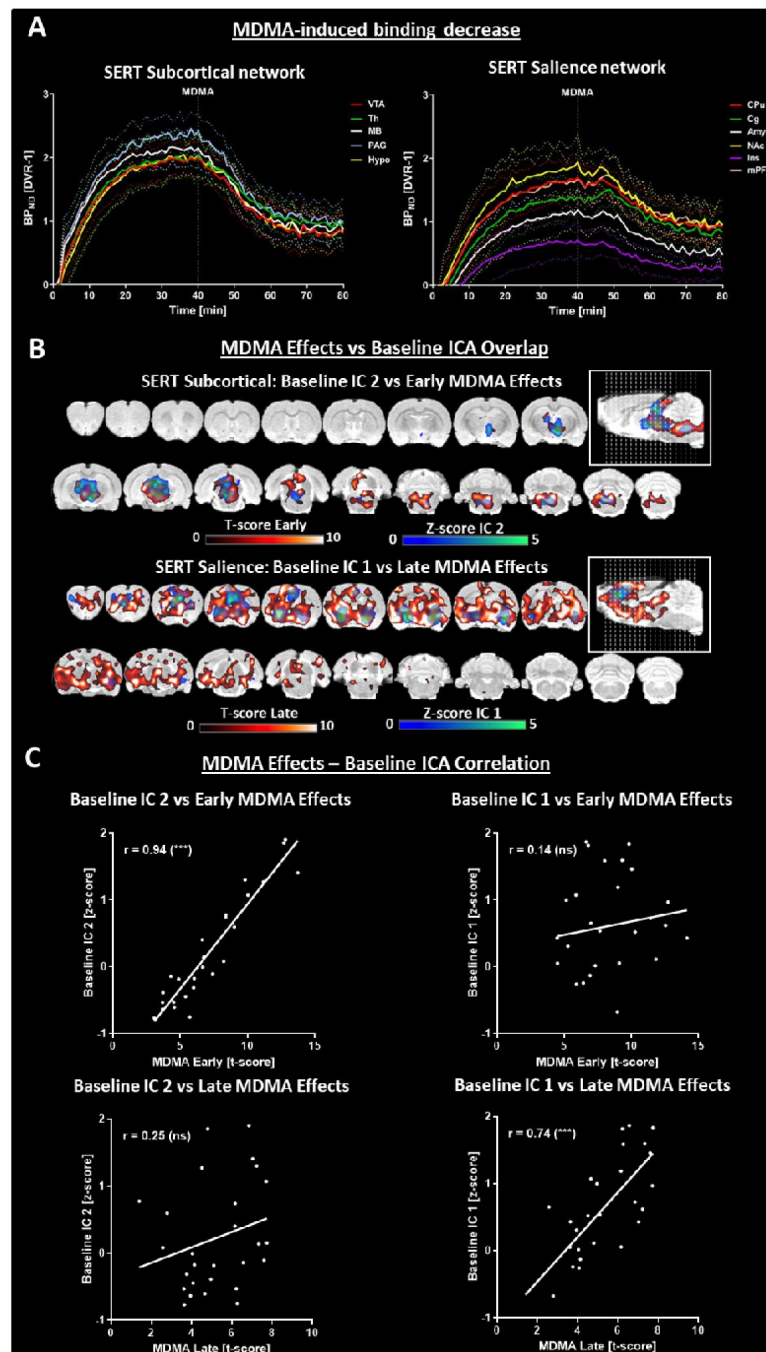


Figure 3

Comparison of MDMA-induced [¹¹C]DASB alterations.

(A) Left panel: Dynamic binding potentials of regions comprising the SERT subcortical network, defined by IC 1 in the validation cohort. Right panel: Dynamic binding potentials of regions comprising the SERT salience network, defined by IC 2 in the validation cohort (continuous lines indicate the means, and dotted lines indicate standard deviations). (B) Overlap between independent components extracted from the validation cohort (IC 1 = SERT subcortical network, IC 2 = SERT salience network;) and the early and late effects of MDMA. (C) Pairwise correlations between regional z scores of the ICs extracted from the validation cohort and regional t scores of early and late MDMA effects. (***) indicates $p < 0.001$, ns = nonsignificant). Abbreviations: SERT = serotonin transporter, ICA = independent component analysis; for abbreviations of regions, please refer to the *Supplementary Information*).

characteristics of [^{11}C]DASB changes after MDMA exposure, we found highly significant correlations between the z scores of the posterior and anterior ICs and the t scores of late and early MDMA effects on [^{11}C]DASB alterations ($p < 0.0001$, **Figure 3C** [↗](#)).

Finally, we investigated the relationships between SERT availability changes and MC reductions following acute MDMA challenge (**Figure 4** [↗](#)).

We found that the decreases in MC and [^{11}C]DASB BP_{ND} following MDMA application (**Figure 4A** [↗](#)) showed only a low correlation ($r = 0.29$, **Figure 4B** [↗](#)). While decreases in SERT availability exhibited a strong anterior-posterior gradient, which was most pronounced in areas with high SERT availability, such as the MB, VTA, Pons or PAG, the MC encompassed regions across the brain to similar extents. Specifically, while the significance of [^{11}C]DASB reductions in Ins was very low across the investigated regions, Ins presented the greatest effects among regions for MC. In contrast, strong SERT occupancy effects in IC and SC did not translate into very prominent reductions in the respective global MCs of the two regions.

MDMA-induced changes in seed-based molecular connectivity

Next, we performed a seed-based analysis to compare changes in FC and MC after an MDMA pharmacological challenge on the salience network and regions with high SERT binding (**Figure 5** [↗](#)).

We observed a decrease in SN FC (16% at the end of the scan, $p = 0.02$, did not survive FDR correction) and almost constant FC of the subcortical network (<5% decrease, $p = 0.79$) following MDMA, as shown in **Figure 5A** [↗](#) at the edge and node levels and **Figure 5C** [↗](#) at the network level. For MCs, we observed profound reductions in the subcortical network (**Figure 5C and D** [↗](#), $p = 0.009$), emphasizing the acute and spatially specific effect of MDMA on MCs.

Discussion

The mammalian brain operates on diverse physiological, spatial and temporal scales. FC via BOLD-fMRI offers insights into coherent functional brain networks, but its complexity and indirect link to neural activity highlight the need for more direct methodologies. In this context, the concept of MC using [^{11}C]DASB PET, as introduced in our study, provides a more direct and complementary perspective on brain organization and its response to external stimuli, such as MDMA.

Physiological basis

Our findings suggest that [^{11}C]DASB binding reflects the interplay of serotonin levels and SERT dynamics [21 [↗](#)]. In support of the competition model, evidence indicates that endogenous serotonin competes with tracers for binding sites, affecting tracer binding [22 [↗](#)]. However, contrasting results from various studies highlight the complexity of this interaction [23 [↗](#)] [24 [↗](#)] [25 [↗](#)]. The internalization model suggests that serotonin levels influence SERT internalization, impacting [^{11}C]DASB binding [21 [↗](#), 26 [↗](#), 27 [↗](#)]. While supporting evidence, further exploration is needed to fully understand these dynamics, especially during the resting state [28 [↗](#)]. Some models on this topic have proposed a regulatory function of the raphe nuclei in maintaining serotonin fluctuations over several temporal scales at rest [29 [↗](#)]. Remarkably, fast microdialysis has resolved multiple spontaneous surges of up to 1500-fold of the basal serotonin occurring during 30-minute intervals [30 [↗](#)]. Additionally, the same study indicated that SERT expression is essential for spontaneous surges and that reduced SERT drastically decreases serotonin spiking. Thus, it is feasible that the correlated temporal fluctuations captured by dynamic [^{11}C]DASB at least partly reflect the role of the serotonergic system in the resting activity of the brain. In addition, SERT regulation occurs over multiple time scales, ranging from milliseconds to hours,

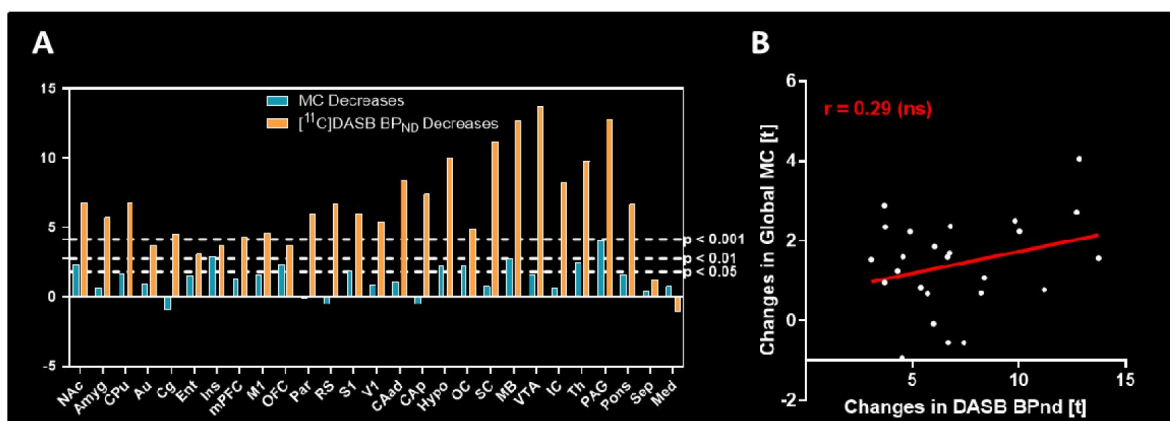


Figure 4

Comparison of MC and BP_{ND} changes following MDMA.

A Reductions in BP_{ND} following MDMA (orange) and MC strength (blue) were compared. **B** The correlation coefficient of the regional T scores was low ($r = 0.29$).

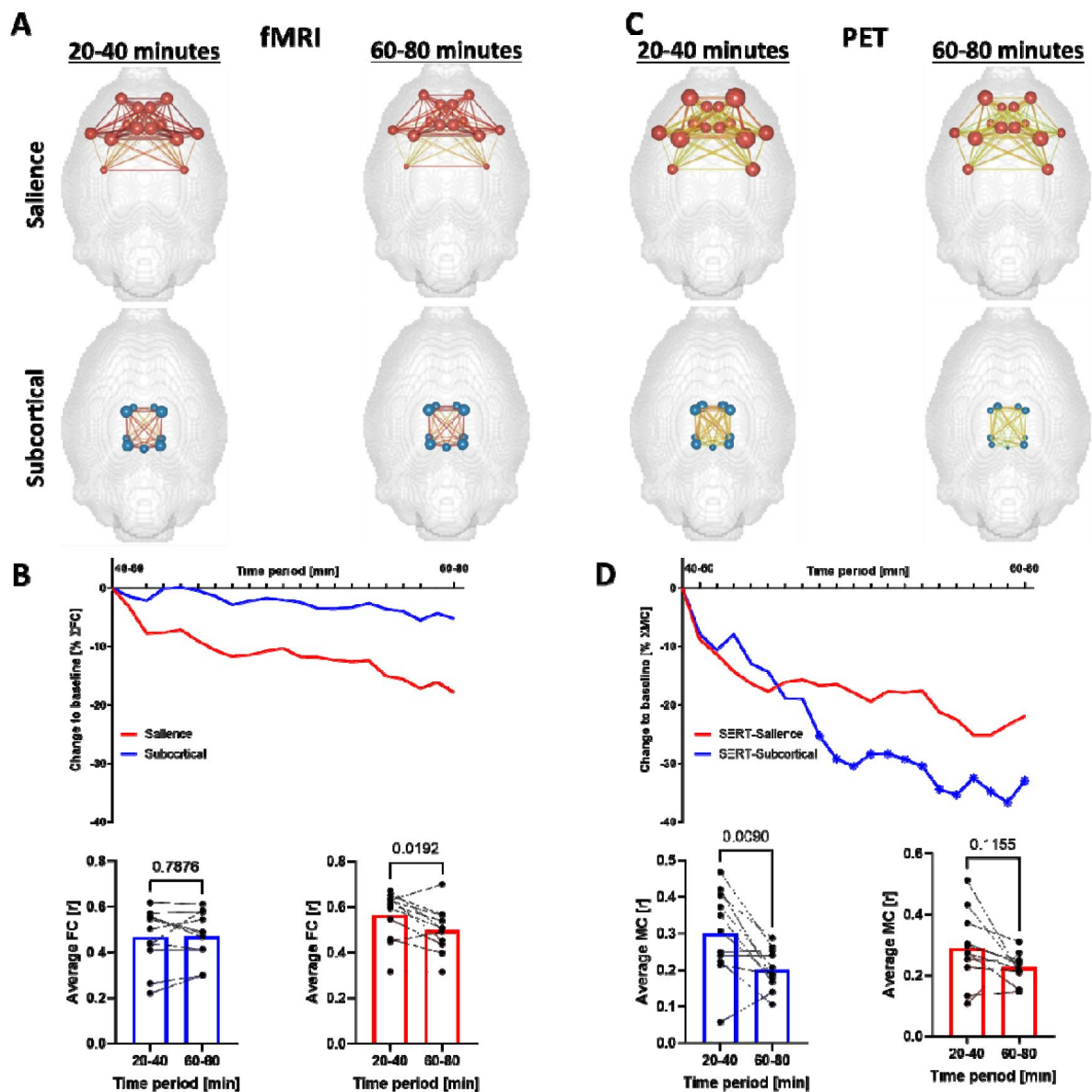


Figure 5

MDMA effects on seed-based FC and MC of the salience and subcortical networks.

(A) FC and (C) MC brain networks depicting the edge and node strengths of the salience network and subcortical network at baseline (20-40 min after the start of the scan) and after MDMA (60-80 min after the start of the scan). (B) FC and (D) MC time-resolved salience and subcortical network strengths computed by sliding windows. Individual values are provided in the bar graphs for the baseline and final post-MDMA periods. Asterisks indicate significant ($p < 0.05$, FDR-corrected) changes from baseline (time point zero, corresponding to 20-40 minutes after the start of the scan).

depending on the mechanism involved [31]. Rapid changes in SERT surface expression can be mediated by protein-protein interactions or posttranslational modifications [32, 33], such as phosphorylation, which occur on a timescale of milliseconds to minutes. These processes dynamically modulate surface availability and function, allowing fine-tuned regulation of serotonin uptake even under resting-state conditions. Additionally, while slower processes involving endocytosis, recycling, and degradation typically occur over minutes to hours, subtle fluctuations in SERT trafficking and activity can still occur under basal conditions. These minor yet biologically relevant changes likely reflect ongoing homeostatic regulation essential for maintaining serotonergic balance. Therefore, tracer fluctuations observed during resting-state measurements should not be dismissed, as they may represent meaningful variations in SERT regulation that contribute to the fine control of serotonin clearance.

Implications for the whole-brain serotonergic system

Our graph theory analysis revealed comparable whole-brain organization between the hemodynamic (BOLD-derived) and serotonergic (PET-derived) connectomes, although we found only a moderate correlation between the edges of the two measurements. Notably, the [^{11}C]DASB-based MC results showed a distinct binding pattern compared with BOLD-fMRI-derived FC. For FC, the ICA revealed classic RSNs, including the default-mode network, sensory network and motor network, consistent with previous studies in rats [34] and humans [35]. In contrast, the PET data revealed two distinct anatomical components in the serotonergic system. One component included key subcortical areas, such as the brainstem, parts of the midbrain and the thalamus, regions rich in SERT availability and central to serotonergic regulation. The other component comprises regions of the limbic system, such as the striatum, amygdala, insula, cingulate cortex and prefrontal cortex, critical targets of serotonergic projections from the raphe nuclei. These components suggest a broader spatial reach of the serotonergic system than the typically described RSN, reflecting its unique neurochemical architecture. In our analysis, PET ICs appeared less bilateral than fMRI ICs. This is likely due to the lower temporal resolution of PET (100 frames) than of fMRI (3000 frames), resulting in a reduced signal-to-noise ratio (SNR) and potentially affecting the stability and symmetry of the independent components.

Our results demonstrate that compared with FC, MDMA induces more pronounced changes in MCs, particularly in regions associated with the SERT subcortical network. The distinct temporal dynamics of BPnd variations between these components may reflect the hierarchical organization of the serotonergic system. Specifically, the raphe nuclei, as the primary source of serotonin, are likely to exert more immediate modulation on posterior subcortical structures (IC2), whereas downstream effects on limbic and cortical regions (IC1) may occur more gradually. While these findings align with current neuroanatomical and molecular knowledge, the precise biological mechanisms driving these temporal differences remain unclear. Future investigations are warranted to elucidate these mechanisms. Future studies combining direct measurements of serotonin levels with neuroimaging data will be critical to fully understanding these components' distinct roles and temporal profiles in regulating serotonergic function.

Relevance to prior research

Our findings align with Salvan et al. [36], who integrated molecular maps into fMRI data and demonstrated how individual serotonergic receptors contribute to network-level activity. Despite our differing methodologies, the receptor activity patterns found, may also correspond to the independent components we found from the [^{11}C]DASB PET data. The dual regression approach reported by Salvan et al. was first reported in the context of pharmacology to delineate the receptor-specific effects of MDMA in humans [37]. The authors found that MDMA specifically decreased FC in the 5HT1A maps in areas that could be ascribed to the human salience network, such as the insula and a collection of medial cortical regions. While not reaching significance, we observed a reduced salience FC and MC in our data.

Previous studies evaluating the effects of MDMA on FC have indicated relatively few effects on FC, as confirmed by the results of this study [38, 39]. Recently, a novel approach [37] revealed decreased connectivity induced by MDMA in areas associated with SERT and 5-HT_{1A} availability. The increased activity in limbic and cortical structures when controlling for vascular effects is accompanied by decreased salience connectivity, indicating that, while neurons become more active through the drug, they do so in an incoherent manner, which would be in line with the hyperactive yet abnormal behavior reported for MDMA abuse. Importantly, the potent vascular effects that play a role in modulating the amplitude of the BOLD-fMRI signal may also influence FC readouts, although the extent of such effects is difficult to estimate [13].

Over the past decade, PET studies using [¹¹C]DASB have focused on the associations of serotonin transporter (SERT) availability across different brain regions, revealing altered interregional SERT connections in patient cohorts, posttreatment changes, and a predictive capacity for treatment response [9, 10, 40] [41] [42]. Our study builds upon this foundation, revealing significant within-subject temporal associations in [¹¹C]DASB binding and demonstrates more pronounced alterations in MC than in FC induced by drugs, thus highlighting the enhanced utility of our method.

Study limitations and future directions

One limitation of our study is that our experimental protocols predate the recently published consensus recommendations for rat fMRI [43], particularly concerning anesthesia and preprocessing pipelines. Using isoflurane anesthesia, although common at the time of data acquisition, introduces a potential confound due to its known neuronal activity effects. However, we previously demonstrated that isoflurane at 1.3% maintains stable physiological parameters and avoids burst suppression [44], a concern at higher doses. Furthermore, other studies have reported that low-dose isoflurane remains feasible for resting-state functional connectivity studies [45]. While isoflurane, a GABA-A agonist, could theoretically interact with the mechanisms of MDMA in the brain, we found no evidence in the literature suggesting significant cross-talk between these substances. Future studies employing medetomidine-based protocols may help minimize this potential confounding factor.

With respect to data preprocessing, we retained the same pipeline used in our prior publications [13, 14] to maintain methodological consistency. While we recognize the advantages of adopting standardized preprocessing, as outlined in the consensus guidelines, this approach ensures comparability with our previous analyses. To facilitate further investigation, we have made the full dataset publicly available (see Data Availability Statement), enabling reanalysis with updated pipelines or additional explorations of this dataset.

Nonetheless, we provide a strong rationale for considering such analyses in future studies. First, we show that at rest, the data are reliable, temporally stable and exhibit similar graph theory metrics to traditionally calculated functional connectomes. Second, despite comparable network-level organizational properties, at rest, MC is only moderately correlated with FC, indicating the complementary nature of both readouts. Third, resting-state MC ICs correlate well with SERT occupancy changes induced by the MDMA challenge. Finally, we show that the changes that MDMA elicits on the MC are complementary to standardly calculated [¹¹C]DASB BP_{ND} alterations. Our data indicate that while changes in BP_{ND} are more pronounced in regions with higher baseline SERT availabilities, MC reveals a more globally distributed measure of tracking serotonergic changes since regions, such as the insula, with relatively low SERT expression, are strongly affected. Additionally, our multimodal imaging approach, although powerful, cannot fully decipher the mechanisms of interregional coherence in PET time courses. Furthermore, studies employing more direct sensitive methods to measure neurotransmitter release could provide deeper insights into the molecular processes underpinning our observations.

Conclusion

This study significantly contributes to integrating molecular data into connectomic frameworks, demonstrating that subject-level MCs are reliable and complementary to FC in both resting and pharmacologically challenged states. Our research provides a strong foundation for future investigations into the value and generalizability of PET-derived MCs, particularly for understanding drug-induced brain-wide molecular network changes.

Additional information

Funding

This research was supported by funds from the Eberhard Karls University Tübingen Faculty of Medicine (fortune 2209-0-0 and 2409-0-0) to HFW, from Bundesministerium für Bildung und Forschung (BMBF, Grant No. 01GQ1415) to HFW and from the Carl Zeiss Foundation to KH.

CRedit authorship contribution statement

Tudor M. Ionescu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Mario Amend:** Investigation, Writing – review & editing. **Rakibul Hafiz:** Methodology, Software, Writing – review & editing. **Bharat B. Biswal:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Andreas Maurer:** Writing – review & editing. **Hans F. Wehrl:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Kristina Herfert:** Writing – review & editing, Visualization, Supervision, Project administration, Funding.

Data availability

The data are openly available at the DRYAD repository (<https://doi.org/10.5061/dryad.6djh9w1bf>).

Additional files

Supplemental methods [↗](#)

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Reviewer #1 (Public review):

This paper by Ionescu et al. applies novel brain connectivity measures based on fMRI and serotonin PET both at baseline and following ecstasy use in rats. There are multiple strengths to this manuscript. First, the use of connectivity measures using temporal correlations of 11C-DASB PET, especially when combined with resting state fMRI, is highly novel and powerful. The effects of ecstasy on molecular connectivity of the serotonin network and salience network are also quite intriguing.

The authors discussed their use of high-dose (1.3%) isoflurane in the context of a recent consensus paper on rat fMRI (Grandjean et al., "A Consensus Protocol for Functional

Connectivity Analysis in the Rat Brain.") which found that medetomidine combined with low dose isoflurane provided optimal control of physiology and fMRI signal. The authors acknowledge their suboptimal anaesthetic regimen, which was chosen before the publication of the consensus paper. This likely explains, in part, why fMRI ICs in figure 2A appear fairly restricted.

The PET ICs appear less bilateral than the fMRI ICs, which the authors attribute to lower SNR.

<https://doi.org/10.7554/eLife.97864.2.sa2>

Reviewer #2 (Public review):

Summary:

The article aims to describe a novel methodology for the study of brain organization, in comparison to fMRI functional connectivity, under rest vs. controlled pharmacological stimulation.

Strengths:

Solid study design with pharmacological stimulation applied to assess the biological significance of functional and (novel) molecular connectivity estimates.

Provides relevant information on the multivariate organization of serotonergic system in the brain.

Provides relevant information on the sensitivity of traditional (univariate PET analysis, fMRI functional connectivity) and novel (molecular connectivity) methods in measuring pharmacological effects on brain function.

Comments on revisions:

I thank the authors for carefully addressing my comments and in particular for the interesting insights added to the discussion.

I have just one last remark pertaining to the point of the sample size: rats undergoing the MDMA acute challenge constitute a relatively small sample (N=11); I feel there is a certain risk the results presented might not be particularly replicable. Could the authors prove the stability of their (main) results by randomly iterating the individuals included in their sample (e.g. via permutation tests)? Alternatively, including at least a justification of the sample size in the context of the available evidence would be valuable.

<https://doi.org/10.7554/eLife.97864.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer 1:

Comment 1- I would like the authors to discuss and justify their use of high-dose (1.3%) isoflurane. A recent consensus paper on rat fMRI (Grandjean et al., "A Consensus Protocol for Functional Connectivity Analysis in the Rat Brain.") found that medetomidine combined with low dose isoflurane provided optimal control of physiology and fMRI signal. To overcome any doubts about the effects of the high-dose anaesthetic I'd encourage the authors to show the results of their functional connectivity specificity

using the same or similar image processing protocol as described in that consensus paper. This is especially true since the fMRI ICs in Figure 2A appear fairly restricted.

We thank the reviewer for their insightful comments. We agree that the combination of medetomidine and isoflurane, as recommended by Grandjean et al. in their consensus paper, provides superior physiological stability and fMRI signal quality, and should indeed be considered the preferred protocol for future studies. In fact, we have adopted this combination in our subsequent research [1]. However, the data acquired in the present study were acquired prior to the publication of the consensus recommendations and have been previously published [2, 3]. While isoflurane is not the ideal anesthetic for functional connectivity studies, we have demonstrated in earlier work [4], that using isoflurane at 1.3% maintains stable physiological parameters and avoids burst suppression, a key issue with higher isoflurane doses.

Regarding preprocessing, we acknowledge the importance of standardized approaches as outlined in the consensus paper. However, to maintain methodological consistency with our prior work, we retained the original preprocessing pipeline for this study. This decision ensures comparability with our previous analyses. To address the reviewer's concerns and encourage further verification, we have uploaded the full dataset to a public repository (as suggested in Comment 4). This will enable other researchers to reanalyze the data using updated preprocessing pipelines or explore additional analyses.

We have updated the manuscript discussion (page 19) to clearly acknowledge these points:

“One limitation of our study is that our experimental protocols predate the recently published consensus recommendations for rat fMRI [42], particularly concerning anesthesia and preprocessing pipelines. The use of isoflurane anesthesia, although common at the time of data acquisition, introduces a potential confound due to its known effects on neuronal activity. However, we previously demonstrated that isoflurane at 1.3% maintains stable physiological parameters and avoids burst suppression [43], a concern at higher doses. Furthermore, other studies have reported that low-dose isoflurane remains feasible for resting-state functional connectivity studies [44]. While isoflurane, as a GABA-A agonist, could theoretically interact with the mechanisms of MDMA in the brain, we found no evidence in the literature suggesting significant cross-talk between these substances. Future studies employing medetomidine-based protocols may help minimize this potential confound.

Regarding data preprocessing, we chose to retain the same pipeline used in our prior publications [13, 14] to maintain methodological consistency. While we recognize the advantages of adopting standardized preprocessing as outlined in the consensus guidelines, this approach ensures comparability with our previous analyses. To facilitate further investigation, we have made the full dataset publicly available (see Data Availability Statement), enabling reanalysis with updated pipelines or additional explorations of this dataset.”

Comment 2 - I'd also be interested to read more about why the cerebellum was chosen as a reference region, given that serotonin is highly expressed in the cerebellum, and what effects the choice of reference region has on their quantification.

This is something we ourselves have examined in a paper, dedicated to determine the most suitable reference region for [11C]DASB, and while the reviewer is correct in saying there is also serotonin in the cerebellum, we found the lowest binding for this tracer in the cerebellar gray matter, recommending this region as a valid reference area. (“Displaceable binding of (11)C-DASB was found in all brain regions of both rats and mice, with the highest binding being in the thalamus and the lowest in the cerebellum. In rats, displaceable binding was largely reduced in the cerebellar cortex”, please refer to [5]).

We amended our materials and methods part to specify that we had shown in this previous publication that the cerebellar gray matter is appropriate as a reference region (page 6):

“Binding potentials were calculated frame-wise for all dynamic PET scans using the DVR-1 (equation 1) to generate regional BPND values with the cerebellar gray matter as a reference region, which our earlier studies have demonstrated to be the most appropriate for this tracer in rats [5, 6]:”

Comment 3 - The PET ICs appear less bilateral than the fMRI ICs. Is that simply a thresholding artefact or is it a real signal?

We thank the reviewer for this observation. The reduced bilaterality of PET ICs compared to fMRI ICs is likely due to the inherent limitation in the temporal resolution of PET, which provides significantly fewer frames (100 frames compared to 3000 frames for fMRI). This lower temporal resolution leads to reduced signal-to-noise ratio when computing the ICA, which can affect the stability and symmetry of the ICs during ICA computation, particularly at higher IC numbers. While thresholding may also a minor role, we believe the primary factor is poorer SNR associated with the PET data. We have clarified this point in the discussion section (page 17) as follows:

“In our analysis, PET ICs appeared less bilateral than fMRI ICs. This is likely due to the lower temporal resolution of PET (100 frames) compared to fMRI (3000 frames), resulting in reduced signal-to-noise ratio (SNR) and potentially affecting the stability and symmetry of the independent components.”

Comment 4 - "The data will be made available upon reasonable request" is not sufficient - please deposit the data in an open repository and link to its location.

We agree with the request of the reviewer and uploaded the data to a Dryad repository. We amended our Data Availability Statement accordingly.

Comment 5 (recommendation) - Please add the age and sex of the rats in lines 92-97.

Amended.

Comment 6 (recommendation) - There are multiple typos throughout the manuscript - for example, "z-vlaue" on line 164, "negligable" on line 194, etc.. Sometimes the 11 in 11C is superscripted, sometimes it isn't. This paper would benefit from a careful proofread.

Thank you for pointing this out. We sent the manuscript for language and grammar editing to AJE (see certificate).

Reviewer 2:

Comment 1 - While the study protocol is referenced in the paper, it would be useful to at least report whether the study uses bolus, constant infusion, or a combination of the two and the duration of the frames chosen for reconstruction. Minimal details on anesthesia should also be reported, clarifying whether an interaction between the pharmacological agent for anesthesia and MDMA can be expected (whole-brain or in specific regions).

We fully agree that this would improve the readability of our manuscript and added the information to the materials and methods and discussion accordingly. Please refer to page 4/5.

Comment 2 - Some terminology is used in a bit unclear way. E.g. "seed-based" usually refers to seed-to-voxel and not ROI-to-ROI analysis, or e.g. it is a bit confusing to have IC1 called SERT network when in fact all ICs derived from DASB data are SERT networks. Perhaps a different wording could be used (IC1 = SERT xxxxx network; IC2= SERT salience network).

Based on the reviewer's suggestion, we suggest to rename IC1 and IC2 according to their anatomical and functional characteristics (page 13):

"IC1 = SERT Salience Network: This name highlights the involvement of the regions typically associated with the salience network (e.g., CPU, Cg, NAc, Amyg, Ins, mPFC), which play key roles in emotional and cognitive processing."

"IC2 = SERT Subcortical Network: This name reflects the involvement of subcortical regions which play a role in arousal, stress response, and autonomic regulation, which are heavily modulated by serotonin in areas like the hypothalamus, PAG, and thalamus."

Comment 3 - The limited sample size for the rats undergoing pharmacological stimulation which might make the study (potentially) not particularly powerful. This could not be a problem if the MDMA effect observed is particularly consistent across rats. Information on inter-individual variability of FC, MC, and BPND could be provided in this regard.

We thank the reviewer for raising this point. To address the concern about limited sample size and inter-individual variability, we have added this information to Figures 5 B and D. Regarding the BPND variability, the dotted lines in Figure 3 indicate the standard deviation in the regional BPNDs, however, this was not clearly stated in the original figure description. We have now amended the figure legend to explicitly clarify this point.

Comment 4 (recommendation) - "Our research employs a novel approach named "molecular connectivity" (MC), which merges the strengths of various imaging methods to offer a comprehensive view of how molecules interact within the brain and affect its function." I'd recommend rephrasing to "...how molecular interact across different areas within the brain...". Molecular connectivity is a potentially ambiguous term (used to study interactions across different molecules (in the same compartment/environment) vs. to study interactions across the same molecules in different areas). I'd add a couple of references to help the reader disambiguate too (e.g. <https://pubmed.ncbi.nlm.nih.gov/30544240/>, <https://pubmed.ncbi.nlm.nih.gov/36621368/>)

We appreciate the reviewer's suggestion and agree that the term "Molecular Connectivity" could be ambiguous. To clarify, we rephrased the description to emphasize that our approach specifically examines interactions of the same molecule (i.e., serotonin transporter) across different brain regions, rather than interactions between different molecules within the same environment. We propose the following revised text (page 2):

"Our research employs a novel approach termed molecular connectivity (MC), which combines the strengths of various imaging methods to provide a comprehensive view of how specific molecules, such as the serotonin transporter, interact across different brain regions and influence brain function."

Additionally, we will incorporate the suggested references to help the reader further contextualize the use of this term.

Comment 5 - In the methods, it is not clear if for MC the authors also compute ROI-to-ROI correlations or only ICA.

Thank you for highlighting this point. To clarify, our MC analysis, includes both ROI-to-ROI correlations and ICA. Specifically, as described at the end of the “Molecular Connectivity Analysis” subchapter, we compute ROI-to-ROI correlations using the following steps: 1. The first 20 minutes of each scan are discarded to account for perfusion effects. 2. A detrending approach is applied to the remaining 60 minutes of BP_{ND} time courses. 3. ROI-to-ROI calculations are then calculated and organized into subject-level correlation matrices, which are subsequently z-transformed to generate mean correlation matrices across subjects.

We revised the methods section to explicitly state that both ROI-to-ROI correlations and ICA are integral components of the MC analysis to ensure this point is clear to readers (page 6).

“The BP_{ND} time courses were then used to calculate MC as described above for fMRI: ROI-to-ROI subject-level correlation matrices between all regional time courses were generated and z-transformed correlation coefficients were used to calculate mean correlation matrices.”

Comment 7 - In the discussion, it could be useful to relate IC1 and IC2 to well-established neuroanatomical/molecular knowledge of the serotonergic system. Did the authors expect the IC1 and IC2 anatomical distributions? is there a plausible biological reason as to why the time courses of BPnd variations would be somehow different between IC1 and IC2?

We appreciate the reviewer’s insightful comment and agree on the importance of relating IC1 and IC2 to well-established neuroanatomical and molecular knowledge of the serotonergic system.

In our discussion, we noted that IC1 primarily encompasses subcortical structures such as the brainstem, midbrain, and thalamus. These regions are consistent with areas housing dense serotonergic projections originating from the raphe nuclei, the primary source of serotonin release. In contrast, IC2 involves limbic and cortical regions - including the striatum, amygdala, cingulate, insular, and prefrontal cortices - which are key targets of the serotonergic pathways. This anatomical distinction aligns with the hierarchical organization of the serotonergic system, where the brainstem nuclei exert both local and distal serotonergic modulation.

The observed differences in the temporal dynamics of the binding potential (BP_{ND}) variations between IC1 and IC2 likely reflect the distinct functional roles of these regions within the serotonergic network. The more immediate changes in IC1 could be attributed to the direct effect of MDMA on the raphe nuclei, leading to rapid serotonin release in subcortical structures. In contrast, the delayed changes in IC2 may reflect downstream modulation in cortical and limbic regions involved in processing more complex emotional and cognitive functions.

That said, while these interpretations are plausible based on current neuroanatomical and functional knowledge, the exact biological mechanisms underlying the differential time courses remain unclear. As discussed in the manuscript, future studies incorporating direct, simultaneous measurements of serotonin levels and imaging data will be essential to fully elucidate the temporal and spatial dynamics of serotonin transmission in these regions. We have revised to better highlight this limitation in the discussion section (page 17) as an important area for further investigation:

“Our results demonstrate that compared with FC, MDMA induces more pronounced changes in MCs, particularly in regions associated with the SERT subcortical network. The distinct

temporal dynamics of BPnd variations between these components may reflect the hierarchical organization of the serotonergic system. Specifically, the raphe nuclei, as the primary source of serotonin, are likely to exert more immediate modulation on posterior subcortical structures (IC2), whereas downstream effects on limbic and cortical regions (IC1) may occur more gradually. While these findings align with current neuroanatomical and molecular knowledge, the precise biological mechanisms driving these temporal differences remain unclear. Future investigations are warranted to elucidate these mechanisms. Future studies combining direct measurements of serotonin levels with neuroimaging data will be critical to fully understanding these components' distinct roles and temporal profiles in regulating serotonergic function."

Comment 8 - In the discussion (physiological basis), could the authors detail the expected "time scale" in changes in SERT expression? How quickly can SERT expression change, especially under resting-state conditions? Is it reasonable to consider tracer fluctuations under rest conditions as biologically meaningful?

SERT regulation can occur over different time scales depending on the mechanism involved [7].

Acute, rapid changes (milliseconds to seconds): Protein-protein interactions with key regulatory proteins (e.g., syntaxin1A, neuronal nitric oxide synthase) can lead to rapid modulation of SERT surface expression [8-11]. These interactions often involve changes in transporter trafficking or conformational states and can occur within milliseconds to seconds. For example, syntaxin1A directly interacts with the N-terminus of SERT, influencing its availability on the plasma membrane within short timescales.

Intermediate time scales (seconds to minutes): Posttranslational modifications, such as phosphorylation by kinases (e.g., protein kinase C) or dephosphorylation by phosphatases, are known to influence SERT function and surface expression [12-14]. These processes are typically initiated in response to cellular signaling and occur over seconds to minutes, affecting the SERT trafficking dynamics and serotonin uptake capacity [15, 16].

Longer-term changes (minutes to hours): Longer-term regulation involves processes like endocytosis, recycling, or degradation of SERT. These pathways typically take minutes to hours and are often part of more sustained cellular responses to changes in neuronal activity or serotonin levels. Such changes are slower but contribute to the overall cellular homeostasis of SERT under prolonged stimulation.

Under resting-state conditions, where neurons are not subjected to rapid or dramatic fluctuations in neurotransmitter release or signaling, SERT expression and activity are generally stable but still subject to subtle fluctuations due to ongoing basal regulatory processes. Basal phosphorylation or low-level protein-protein interactions can still dynamically modulate SERT trafficking and function, albeit at a lower intensity than under stimulated conditions. These fluctuations, although smaller in magnitude, may reflect fine-tuning of serotonin homeostasis and can occur on shorter timescales (seconds to minutes).

Biological Relevance of Tracer Fluctuations at Rest:

It is reasonable to consider that tracer fluctuations under resting conditions could reflect biologically meaningful variations in SERT expression and function. Even subtle shifts in SERT surface availability or activity can impact serotonin clearance and signaling, given the fine balance required to maintain serotonergic tone. These fluctuations may reflect intrinsic neuronal variability or ongoing homeostatic adjustments to maintain optimal neurotransmitter levels or serve as early indicators of adaptive responses to environmental or physiological changes before more overt modifications in transporter expression or activity become apparent.

In summary, while SERT expression can change rapidly in response to signaling events (milliseconds to minutes), even under resting-state conditions, subtle regulatory fluctuations can be biologically meaningful. These fluctuations likely reflect ongoing regulatory adjustments essential for maintaining serotonergic balance and should not be disregarded as noise, particularly in experimental measurements using tracers.

We added the following paragraph to the discussion (page 16):

“In addition, SERT regulation occurs over multiple time scales, ranging from milliseconds to hours, depending on the mechanism involved [31]. Rapid changes in SERT surface expression can be mediated by protein-protein interactions or posttranslational modifications [32, 33], such as phosphorylation, which occur on a timescale of milliseconds to minutes. These processes dynamically modulate surface availability and function, allowing fine-tuned regulation of serotonin uptake even under resting-state conditions. Additionally, while slower processes involving endocytosis, recycling, and degradation typically occur over minutes to hours, subtle fluctuations in SERT trafficking and activity can still occur under basal conditions. These minor yet biologically relevant changes likely reflect ongoing homeostatic regulation essential for maintaining serotonergic balance. Therefore, tracer fluctuations observed during resting-state measurements should not be dismissed, as they may represent meaningful variations in SERT regulation that contribute to the fine control of serotonin clearance.”

Comment 9 - In the discussion, the SERT network results should be commented on more extensively, as there is now only a generic reference to MC changes being stronger than FC ones, without spatial reference to the SERT network (while only negative salience network results are referenced explicitly instead, making the paragraph a bit confusing).

We expanded the discussion to accommodate a more thorough contemplation of this network. This revised paragraph (page 17) directly addresses the spatial aspects of the SERT network, highlighting the specific regions involved in serotonergic connectivity and contrasting molecular and functional connectivity changes induced by MDMA.

Comment 10 - Figure 3; I'd switch left and right charts in the bottom panel (last row only), to keep the SERT network always on the left of the Figure.

We agree with the suggestion and changed the figure accordingly.

Comment 11 - Figure 4: I'd add FC decreases to the figure, to allow the reader to compare BPnd, MC, and FC changes more easily and I'd add a horizontal line at the equivalent of e.g. Z-1.96 (or similar) so that it is clear which measures/regions display significant changes.

We prefer to keep the figure focusing on the two analyses of PET alterations, since we want to emphasize their complementarity in the context of PET specifically. However, we added lines indicating significances, in line with the reviewer's suggestion.

Comment 12 - In Figure 5D, the y-axis mentioned FC but I suppose it should mention MC.

We amended the figure accordingly, together with the changes to the names of the networks implemented across the manuscript.

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<https://doi.org/10.7554/eLife.97864.2.sa0>